

A Compact Block-Diagonal Subcase of Tracial Joint Spectral Measures

FA/Banach automatic math run

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Source And Target

The source is Otte Heinavaara, *Tracial joint spectral measures*, arXiv:2310.03227. For Hermitian matrices A, B , the source constructs a positive measure $\mu_{A,B}$ on $\mathbb{R}^2$ satisfying

$$\mathrm{tr} H(f)(xA + yB) = \int_{\mathbb{R}^2} f(ax + by) d\mu_{A,B}(a, b), \quad (1)$$

where

$$H(f)(s) = \int_0^1 \frac{1-t}{t} f(ts) dt.$$

Section 1.8 of the source lists as future work the “existence and uniqueness of the measures $\mu_{A,B}$ for compact self-adjoint operators A and B .”

1.8. Future work. Our structural result opens interesting research directions. Some of these will be pursued in our forthcoming work [\[Hei23\]](#), which in particular will study the following topics:

- Geometric and regularity properties of μ_c, μ_s and their supports.
- Further consequences and reformulations of the main identity [\(3\)](#).
- The structure of measures $\mu_{A,B}$ for small matrices A and B .
- Existence and uniqueness of the measures $\mu_{A,B}$ for compact self-adjoint operators A and B .
- Optimality results which show various ways in which our main result cannot be improved.
- Establishing a relationship to hyperbolic polynomials in the sense of Gårding [\[Gär59\]](#).

Result

We prove a compact-operator subcase: common finite-dimensional reducing blocks with square-summable Hilbert-Schmidt size. This includes the important commuting compact Hilbert-Schmidt case, by simultaneous diagonalization.

Idea Of The Proof

The finite-dimensional measure is already available on each reducing block. Because the blocks are square-summable in Hilbert-Schmidt norm, the block measures have summable second moments. This gives local finiteness away from the origin, so their countable sum is a Radon measure on the punctured plane. The source identity then sums block by block for tests supported away from zero. Uniqueness follows from the same moment-determinacy mechanism used in the source:

after multiplying by $a^2 + b^2$, one obtains a finite compactly supported measure whose polynomial moments are fixed.

Theorem 1 (Block compact tracial joint spectral measures). *Let $\mathcal{H} = \bigoplus_{j=1}^{\infty} \mathcal{H}_j$ be an orthogonal direct sum of finite-dimensional Hilbert spaces. Let*

$$A = \bigoplus_{j=1}^{\infty} A_j, \quad B = \bigoplus_{j=1}^{\infty} B_j$$

be self-adjoint block-diagonal compact operators, where $A_j, B_j \in B(\mathcal{H}_j)$ are Hermitian matrices, and assume

$$\sum_{j=1}^{\infty} (\operatorname{tr} A_j^2 + \operatorname{tr} B_j^2) < \infty. \quad (2)$$

For each j , let $\mu_j = \mu_{A_j, B_j}$ be the finite-dimensional tracial joint spectral measure of the source. Then

$$\mu_{A, B} := \sum_{j=1}^{\infty} \mu_j$$

defines a positive Radon measure on $\mathbb{R}^2 \setminus \{0\}$, supported in $\{(a, b) : |a| + |b| \leq \|A\| + \|B\|\}$, with finite second moment

$$\int_{\mathbb{R}^2 \setminus \{0\}} (a^2 + b^2) d\mu_{A, B}(a, b) = \frac{1}{6} \sum_{j=1}^{\infty} (\operatorname{tr} A_j^2 + \operatorname{tr} B_j^2). \quad (3)$$

Moreover, for every bounded Borel $f : \mathbb{R} \rightarrow \mathbb{R}$ whose support is compact and does not contain 0, and every $x, y \in \mathbb{R}$,

$$\operatorname{tr} H(f)(xA + yB) = \int_{\mathbb{R}^2 \setminus \{0\}} f(ax + by) d\mu_{A, B}(a, b). \quad (4)$$

For every $p \geq 2$,

$$\operatorname{tr} |xA + yB|^p = p(p+1) \int_{\mathbb{R}^2 \setminus \{0\}} |ax + by|^p d\mu_{A, B}(a, b). \quad (5)$$

Finally, $\mu_{A, B}$ is unique among positive Radon measures on $\mathbb{R}^2 \setminus \{0\}$ with bounded support and finite second moment satisfying (4) and the polynomial identities obtained from (5) for integer $p \geq 2$.

Proof. For a finite-dimensional Hermitian pair (C, D) , apply the source identity with $f(t) = t^2$. Since $H(f)(t) = t^2/6$, polarization gives

$$\int_{\mathbb{R}^2} (ax + by)^2 d\mu_{C, D}(a, b) = \frac{1}{6} \operatorname{tr}(xC + yD)^2.$$

Taking $(x, y) = (1, 0), (0, 1)$ and adding yields

$$\int_{\mathbb{R}^2} (a^2 + b^2) d\mu_{C, D}(a, b) = \frac{1}{6} (\operatorname{tr} C^2 + \operatorname{tr} D^2). \quad (6)$$

The source uniqueness proof also gives the support bound $\operatorname{supp} \mu_{C, D} \subseteq \{(a, b) : |a| + |b| \leq \|C\| + \|D\|\}$.

Let $K \subset \mathbb{R}^2 \setminus \{0\}$ be compact and put $\delta = \text{dist}(K, 0) > 0$. By (6),

$$\sum_j \mu_j(K) \leq \delta^{-2} \sum_j \int_K (a^2 + b^2) d\mu_j \leq \frac{1}{6\delta^2} \sum_j (\text{tr } A_j^2 + \text{tr } B_j^2) < \infty.$$

Thus $\sum_j \mu_j$ is locally finite on the punctured plane. The same calculation proves the second-moment formula (3). The support bound follows from $\|A_j\| \leq \|A\|$ and $\|B_j\| \leq \|B\|$.

Now let f be bounded and supported in a compact subset of $\mathbb{R} \setminus \{0\}$. For fixed $(x, y) \neq (0, 0)$, the function $(a, b) \mapsto f(ax + by)$ is supported in a closed subset of the bounded support of $\mu_{A,B}$ that stays a positive distance from the origin. Hence the right side of (4) is the absolutely convergent sum of the block integrals. On each block, the source identity gives

$$\text{tr } H(f)(xA_j + yB_j) = \int_{\mathbb{R}^2} f(ax + by) d\mu_j(a, b).$$

Summing in j proves (4); the absolute convergence also shows that $H(f)(xA + yB)$ is trace class. The case $(x, y) = (0, 0)$ is immediate.

For $p \geq 2$, finite-dimensional source identity with $f(t) = |t|^p$ gives the block formula. Since

$$\text{tr } |xA_j + yB_j|^p \leq \|xA + yB\|^{p-2} \text{tr } (xA_j + yB_j)^2$$

and the right side is summable by (2), the block traces sum absolutely. Summing the block identities gives (5).

It remains to justify uniqueness. Suppose ν is another positive Radon measure on $\mathbb{R}^2 \setminus \{0\}$ with bounded support and finite second moment, satisfying the same polynomial identities. Define finite compactly supported measures

$$d\tilde{\mu} = (a^2 + b^2) d\mu_{A,B}, \quad d\tilde{\nu} = (a^2 + b^2) d\nu.$$

For every homogeneous polynomial q of degree at least 2, polarization of the identities for $(ax + by)^m$ gives

$$\int q(a, b) d\mu_{A,B} = \int q(a, b) d\nu.$$

Consequently, for every polynomial $r(a, b)$,

$$\int r(a, b) d\tilde{\mu} = \int r(a, b)(a^2 + b^2) d\mu_{A,B} = \int r(a, b)(a^2 + b^2) d\nu = \int r(a, b) d\tilde{\nu}.$$

Compactly supported finite Borel measures on $\mathbb{R}^2$ are determined by their polynomial moments, so $\tilde{\mu} = \tilde{\nu}$. Dividing by $a^2 + b^2$ on the punctured plane gives $\mu_{A,B} = \nu$. $\square$

Corollary 1 (Commuting compact Hilbert-Schmidt pairs). *Let A and B be commuting compact self-adjoint Hilbert-Schmidt operators on a separable Hilbert space. Then A and B admit a unique tracial joint spectral measure in the sense of Theorem 1.*

Proof. Commuting compact self-adjoint operators are simultaneously diagonalizable: there is an orthonormal basis in which $A = \text{diag}(a_j)$ and $B = \text{diag}(b_j)$. The Hilbert-Schmidt assumption says $\sum_j (a_j^2 + b_j^2) < \infty$, so Theorem 1 applies with one-dimensional blocks. In this case the measure is explicit:

$$\mu_{A,B} = \sum_j (t \mapsto t(a_j, b_j))_{\#} \left(\frac{1-t}{t} dt \Big|_{(0,1]} \right).$$

$\square$

Verification Notes

The script `code/verify_commuting_model.py` checks the normalization in Corollary 1. For a diagonal Hilbert-Schmidt model it verifies

$$\int_0^1 \frac{1-t}{t} |t(xa_j + yb_j)|^p dt = \frac{|xa_j + yb_j|^p}{p(p+1)}$$

and prints the finite second-moment matrix.

Limitations And Novelty Check

This is a partial result. It does not prove existence for arbitrary compact self-adjoint pairs. A natural compression proof for all Hilbert-Schmidt pairs would require a continuity/tightness theorem for finite-dimensional tracial joint spectral measures that prevents second-moment loss at the origin; that step is not proved here.

Cheap run indexes were searched for arXiv:2310.03227 and for tracial joint spectral measure keywords. The source bibliography lists the follow-up “Properties of Tracial Joint Spectral Measures” as in preparation, and a bounded web search for that title and for compact self-adjoint tracial joint spectral measures did not find an available arXiv answer.

References

References

- [1] O. Heinavaara, *Tracial joint spectral measures*, arXiv:2310.03227, 2023.